SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH356

Examination - July Semester 2023

Advanced Mathematical Optimisation

Tuesday, 21 November 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SIX (6)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Define the function $f : [-1, 1] \longrightarrow \mathbb{R}$ by

$$f(x) = x - \frac{3}{4}\sin x - \frac{1}{4}.$$

Question 1a

Show that the function f has at least one root.

(4 marks)

Question 1b

Assume we are going to use the Simple Iteration Method to find a root of $f(x)$ in the interval $[-1, 1]$. Analyse whether the Simple Iteration Method is convergent using the iteration formula

$$x = g(x) = \frac{1}{4} + \frac{3}{4}\sin x.$$

(7 marks)

Question 2a

Define the function $f : (0, +\infty) \longrightarrow \mathbb{R}$ by

$$f(x) = x \ln x + x + \frac{1}{x}.$$

Show that the function is convex on the interval $(0, +\infty)$.

(6 marks)

Question 2b

If $f(x)$ and $g(x)$ are two convex functions on $\mathbb{R}$, show that the function

$$h(x) = \max\{f(x), g(x)\}$$

is also a convex function.

(6 marks)

Question 2c

Determine whether the set $S = \{(x, y) \in \mathbb{R}^2 | x^2 + 2x + y^2 - 4y - 8 \leq 0\}$ is convex.

(8 marks)

Question 3a

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$

$$f(x) = x^2 - e^x + 2.$$

Solve the function $f(x) = 0$ for three iterations using Newton's method by starting from the initial point $x_0 = 0$. Express your answer correct to three decimal places.

(8 marks)

Question 3b

Define the function $g : (0, +\infty) \longrightarrow \mathbb{R}$

$$g(x) = x - \ln x - 2.$$

Solve the function $g(x) = 0$ for three iterations using Add-nx method by starting from the initial point $x_0 = 4$. Express your answer correct to three decimal places.

(8 marks)

Question 4a

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x, y, z) = e^x - x + y^2 - 2y + z^2 - \cos z.$$

Find all the stationary points of the function f and determine whether they are local minima, local maxima, or saddle points.

(8 marks)

Question 4b

Define the function $g : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$g(x, y) = e^x - xy + y^2 .$$

Apply one iteration of Newton's method to minimize the function g by using $(2, 2)^T$ as the start point. Find the value of $\left(x^{(1)}, y^{(1)}\right)$ and hence $f\left(x^{(1)}, y^{(1)}\right)$.

(10 marks)

Question 5

Consider the following nonlinear programming problem (NLP).

$$\begin{aligned} \min f(x) &= -\ln(x_1 + 1) - x_2 \\ \text{s.t. } \ 2x_1 + x_2 &\leq 4 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned}$$

Question 5a

Show that all feasible points are regular.

(8 marks)

Question 5b

Solve all the points that satisfy the Karush-Kuhn-Tucker (KKT) conditions for the problem.

(11 marks)

Question 5c

Determine whether the NLP has a global minimizer. Solve the global minimizer if it exists and calculate the function value at the point.

(6 marks)

Question 6

Consider the following NLP,

$$\begin{aligned} &\min x^2 + y^2 \\ &s.t. \quad x + y \geq 1 \end{aligned}$$

Write the Lagrange dual problem and then solve the optimal solution of it.

(10 marks)

----- END OF PAPER -----